

# ActInf GuestStream 084.1 ~ Anthropocentric bias and the possibility of artificial cognition

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hello and welcome everyone this is active inference guest stream 84.1 on July 22nd 2024 on anthropocentric bias and the possibility of artificial cognition with rafhael M and Charles ratov so Raphael and Charles thank you very much for joining to you for introductions and to take us through the paper and welcome everyone this is stream 84 the YouTube July 22nd 2024 thank you okay thank you guys go for it hi thanks for having us so I'm Rafael I'm uh assistant professor at MC University in Australia Sydney in Charles I'm Charles ratov I'm a permanent research associate at the ulish research center in ulish Germany uh in a big Neuroscience Institute so should we go through the paper briefly so um so yeah we wrote this paper uh together actually we started by writing the general audience piece that was published in box online um that was when was that few months ago I guess yeah I think almost yeah close to a year ago maybe I don't think it was published a year ago I think it was published in 2022 but yeah we we worked in for while um yeah so this this piece um was doing um I guess two things that piece of fish in box he was pushing back against what we call the All or Nothing principle um which we Define as the idea that either um something has mind or it doesn't so this kind of neat but uh overly simplistic perhaps partition of things into minded and non-minded things and we argue that this was uh not the best framing to think of uh especially unfamiliar systems that seem to have sophisticated behavioral capacities like large language models or AI systems in general where we don't want to take various cognitive capacities as a package and package them into this idea of a mind where either you have the mind or you haven't and if you have the mind you have all of these things as a package Consciousness reasoning uh Etc uh planning memory um theory of Minds so we thought so as a remedy to to this kind of all and nothing approach we uh argued for a more what we call a divide and conquer strategy when studying the cognitive capacities of of these systems which involve um looking at these capacities on a Pac me basis Case by case with an open-minded empirical approach um yeah child I know if you have anything else that's on the V piece of the

backgrounds yeah I mean we made one point in there about um why it is that people feel so torn about reactions to large language models and we said a little bit about the psychology of essentialism uh which is the idea that um we naturally categorize especially living things with respect to a presumed Essence um so we gave an example of an oak tree I think um and we said that what people tend to think as they grow up and learn about the natural world is that uh an oak tree remains an oak tree regardless of uh changes to its observable properties and um what makes it an oak tree is this unobservable essence of oak or whatever that it presumably has and um there's some experimental psychology and developmental psychology showing that we have a similar attitude towards mindedness or having a mind uh and that is a somewhat speculative explanation for why the literature on large language models is so torn and some people are quite dismissive and other people think that it's um a step away from AGI uh it's that if you feel like you've got to put large language models into one of two boxes the box that you know has the essence of mindedness or the box that lacks it then um you will be forced either to say it doesn't have what it takes to do any of the things that we associate with having a mind such as reasoning or um it has the essential characteristics of mindedness and therefore we should expect it to have all of the other properties we associate with mindedness as well such as Consciousness or um understanding or right I think it's worth yeah emphasizing as you did that the the the background motivation for starting to write on this General piece in the first place is indeed that the the general discourse on AI systems and llms in particular is extremely polarized in a way that is that is very dichotomous and Stark so you have on the one hand people on one hand people um arguing that these systems are no no more than um so-called stochastic parrots that are halfly stitching together samples on the training data and Regita them uh or that there are no smarter than toaster or that you know they only do next token prediction on export prediction and therefore it is um uh a non-starter to ascribe to them any form of cognitive capacity or maybe even uh category mistake uh and on the other hand of the spectrum on the other end of the spectrum you have people um arguing that the systems have D of superhuman intelligence that they exhibit Sparks of artificial general intelligence to the or literally it's literally the title of a of a paper by Microsoft on4 um and many people hyping up the capacity of the systems in a way that might seem very speculative and untethered from actual empirical results so there is this huge gap between these two positions and there's person be a very rich and complex Nuance middle ground that is underexplored um part partly perhaps I think we we did make that point if not explicitly in the published piece and in some draft that um there's something reassuring about being able to make definitive

claims about what these systems are and what they do right so either they're really unsophisticated or they're really very much like us and uh either of these claims kind of meet somewhere in in a way uh in saying that we we have a clear idea of what the systems do and what they are and I think it's a little more epistemic and comfortable to say we we have to um study them empirically and find out what they can or cannot do and why uh and what are the underlying mechanisms and we simply don't know a priori just by looking at the architecture the the learning objective the training data these sources of evidence are insufficient to make the definitive claims about what these systems are capable of so I think that's that was part of the a big part of the motivation and that feeds into that uh more um academic paper as well yeah one other small side note which we don't make in the paper but I think might be relevant um especially for people working in philosophy uh llms are epistemically uncomfortable I think that was the phrase you just used ref which is fitting um not only because they're so new and different but also because they are artifacts right they're things that humans have constructed engineered and um we don't have a thorough understanding of how they work I mean mechanistic interpretability and various behavioral uh research is helping us improve our understanding but on the whole our understanding is not nearly as deep as hopefully it one day will be and this is by itself a really strange situation to be in that we've constructed an artifact that we only partially understand in the um in the past artificial intelligence was seen as um a way of constructing something like an epistemic assistant right something that will help us but not something that will kind of alienate us from the process of coming to know about the world so I think there's an extra layer of discomfort um built into thinking about large language models and that may also play into the [Music] um the uh divisiveness of debates about what they can do and and just to add to that um I guess we should be clear that this does not entail in any way that we think the systems are so completely alien and Beyond the rich of our current understanding that anything goes and that they could very well be uh you know uh have like human-like intelligence or superhuman intelligence and we simply cannot say whether or not they do or because that's sometimes what you see in some opets where people frame these systems as novel alien forms of intelligence that we have created but do not understand or control and that is you know is a slippery slope that leads some people to then claim that they have all these qu magical uh abilities and that's not all what we want to say here and in fact we want to resist that's a c out yeah yeah yeah yeah so we we we think that that's just as much of a cup out as um completely U dismissing a priori what these systems might be capable of without doing the work of

looking into the capacities with behavior and studies so we very much want to resist both extremes of the spectrum if that makes sense okay so now should we move towards the uh content of the current paper sounds good yeah okay may maybe rapael I just start with the distinction between anthropomorphism and anthropocentrism and then you can take the next step um so everyone is aware of the problem of anthropomorphic bias in some form I mean anthropomorphism is just the idea of projecting human qualities onto something nonhuman and it's quite easy to especially when you're having a productive successful exchange with a large language model um it's easy to slip into this interpretive mode where you reason about the responses of the large language model uh as if they were coming from an agent just like you and um maybe that's a useful thing to do in some circumstances but uh from a theoretical perspective it's certainly a mistake because large language model is uh radically unlike you know a human agent in all sorts of ways but um that's only one form of sort of human Centric bias the other one is um anthropocentrism or what we call them Vox article anthropocentric chauvinism and that idea is pretty straightforward it's um the idea that the human way of solving problems is the gold standard of solving problems generally so that to the extent that a system solves a problem in a way that diverges from the human strategy it's uh only using a trick or a fox solution um it's uh it's not using a deep General rational strategy um and in the debate about what large language models can do we think that the anthropomorphic bias is pretty well recognized in the anthropocentric bias is not so well recognized and so part of this paper is is or the main idea behind the paper is to um present a systematic analysis of uh anthropocentric bi how it comes about and how to push back against it right and and we want to be very clear and hopefully we're clearing the paper that the reason why we focus on anthropocentric bias here is just because it is I think as Charles mentioned less um less discussed um and less recognized or some forms of it I recognized and we make we propose this new taxonomy um but it's not at all to suggest that it's uh more problematic or more important than the anthropomorphic bias that's well discussed in the literature of the anthropomorphic biases plural um so in other words this is not um you know to to to frame things in in this slightly problematic dichotomous way of thinking that's common in the discourse on llm this is not a paper that is Ping to the llm booster or llm hype Camp um even though it is pushing back against a certain form of dismissal of authentic biases that only exclusively emphasizes theomorphic biases um but perhaps we should flesh this out a little bit already with the first point we make here in the paper about the performance competence distinction which is a nice for to bring about both anthropomorphism and anthropocentrism regarding um

ATMs so this distinction uh is a very classic distinction in linguistics and cognitive science and it has already been applied to AI systems the networks productively like people like chess Farson um so there's nothing really new here but the distinction comes from no chumsky originally and the idea is that um performance pertains to the external behavior of a system um in a particular domain and uh competence is the kind of set of underlying knowledge and computations U or mechanisms that uh enable the system to achieve that behavior um and a familiar observation in in linguistics Linguistics and cognitive science is that there is a double dissociation between performance and competence so if you take uh for example um language um I might during this very podcast make some grammatic mistakes or some other mistakes in fact I've already um misspoken a few times I think and repeated myself so I've made performance errors um but this does not entail necessarily hopefully that I'm an incompetent language user and that I don't have the line linguistic competence um so that's a well recognized dissociation you can be competent and yet make some errors um and the reason for that is that there might be some additional factors that are unrelated to the underly competence that might impede on my performance so for example I might be distracted when I speak or there might be other um effects uh on my speaking performance that don't actually originate from a lack of competence but just impede on my on on the the the the idealized expression external manifestation of my competence and this is why I miss speak but it's also what recognize that you can have good performance without competence so we give here the example of a student chating on a test or memorizing Test Answers by brute forcing U the test it's a slightly more um I guess gray area but at least in the cheating case a student can a a test without being competent in whatever the test is actually testing for um and it's also well acknowledged in cognitive science that they can do instances like this throughout um you know um like that that can be manifest certain experimental settings where the test subject is right for the wrong reasons as it were NE is doing well it's exhibiting good performance and then we realize the underlying reason for good performance is that there was some perhaps some curis stic that the experimenters or the scientists having thought about that could for this good performance but but doesn't actually amount to whatever competence that we're setting out to test um so we we start by saying well this is what we recognize in linguistics this is the association that suppli to humans across the board you can have performance without competence good performance without competence and you can have bad performance despite competence now when it comes to llms the point we make just going to scroll as we go um we have some figures to sh later but the point we make is that um generally people stress the

dissociation apply the distinction to LMS but stress the sociation only in One Direction unlike in the case of human where it's bidirectional and so what people do generally is to say well LMS famously you know if you look at the gp4 technical report and various other any any report about a new state of the LM uh they are getting really really good add a number of tests and number of benchmarks and even um human exam examinations human exams uh the bar exam medical exams Etc so they can they really good performance uh tests that we tend to think are really difficult tests that one can only uh pass at least a human could only pass if they have a really significant non amount of competence in C domain um and the point that is made when it comes to lms is be careful slow down and try to find out why the model is doing well on that test because there are various reason why it could do well that do not actually indicate that the model has the underlying competence that a test was designed for when it comes to humans so one big concern for example is data contamination where very large language models TR internal scale data um uh can easily be trained on some test items from common benchmarks that leak into the training data such that they can then do really well on the on the on the Benchmark just because they've essentially memorized um test items and there are other more subtle um more subtle reasons why um performance could be very good for the wrong reasons so that's very well recognized and a lot of the people who push back against anthropomorphic bias when it comes to LM make that point careful do not take onic attitude to the systems the reason why they do well is not because they have human like intelligence or human like cognitive capacities but it's for this trivial contingent or um you know otherwise relevant reasons um that accom for the good performance now when it comes to the other dissociation ofation the other direction their people are very reluctant to apply to LMS and we think it's because it's essentially people think in a human case you can make sense of the idea that a human could do badly on a test or could perform badly on a task and yet have the competence that you're trying to test but there might be some auxiliary factors such as um working memory limitations um attention deficits Etc that could impeded on the performance but in the language model I I think we what what we argue in the in the paper we want to argue in the longer version of the paper is that um people don't a lot of people don't think that there is an analogous um mechanism at play where there could be some kind of oxil factor that impedes on performance performance is what you get what you see is what you get and and and so the performance is a direct manifestation of um what the system is computing if if performance if if you have performance errors that can only be explained by the lack of competence because there is no additional um uh independent factor or module that could

impede on the performance sources of interference you might say yeah great yeah um so yeah I mean I'll I'll um pass it over to you Charles I just wanted to to set up the this distinction yeah yeah um right so I mean if you think about a traditional uh computer program at least if you think about a simple computer program it's odd to think of it as some sort of complex systems where complex system where one part of it could sort of interfere with the workings of another part of it but um one of the points we want to make is that um something like that is a realistic um possibility with large language models um okay but I I suppose the next part of the paper goes into a taxonomy of anthropocentric bias and the first sort of overarching point is the distinction between type one and type two so the type one anthropocentrism is a tendency to assume that lm's performance failures um designed to measure competence um always indicates that it lacks uh that competence and so we I before we so we'll say something about three different kinds of type one anthropocentric bias but first a a background Point um which is that um whenever we think it's possible to give a mechan istic explanation of some complicated phenomenon we always have to foreground some factors some variables uh and background others and um the properties that we push into the background um nevertheless matter we're still making assumptions about the nature of those properties when we try to articulate what's going on with the other properties that we're paying more attention to and if assumptions about those properties in the background turn out to be wrong then those mistakes will um corrupt our explanation that attends only to the foregrounded factor so that's a little bit abstract let me just give you a simple example um in comparative cognition one famous uh behavioral experiment is the mirror test for self-recognition so the question is roughly um do non-human animals have something like uh a concept of self and the strategy is to um put some sort of Mark on their body uh and the original experiments it was a Red Dot on the forehead of uh maybe a monkey and or a bird or whatever and then um you put that animal in front of a mirror and see if it um makes any attempt to get rid of the mark and if it does make an attempt to get rid of the mark that shows that it recognizes that the image in the mirror is an image of itself and otherwise not um so that that's a a cool way to get it a really difficult and Abstract question about the mind of a non-human animal but it presumes or it assumes that animals will care about the fact that they have a red dot on their forehead that they will be bothered by that and be motivated to get rid of it and if that assumption is wrong then they might fail the mirror test for self-recognition um for reasons that have little to do with the presence or absence of a capacity for self-recognition and so something similar to that is going on uh we say in large language models um so the first uh example that we give is to do with

task demands so uh you can it's it's a pretty natural uh idea that when Whenever you set up a behavioral task um there will be demands associated with that task that are not directly related to the capacity that you're trying to test so to take the most obvious example that I can think of if you give someone a written test they have to be able to write they have to you know have a hand and a pen and whatever uh and um if their you know hand uh was injured or whatever and they couldn't write then um their failure to fill out the test wouldn't tell you anything about their uh you know academic knowledge um so we suggest that there are auxiliary task demands uh in behavioral tests of a large language model and they're subtle um you wouldn't you wouldn't think of this right away but um we talk about a paper from uh who and Frank uh they have a couple papers on on this topic but what they do is they give a large language model um the following sort of question uh this is a question it's for a grammaticality judgment so you can see on the image there which sentence is better in English number one every child has studied number two every child have studied uh answer with one or two and it gives uh the wrong output um but then you can also simply look at the probabilities assigned to each of those sentences within the model and uh figure out directly whether the model thinks that input a is more likely than input B and it turns out that on a wide variety of questions of this kind um the direct comparison uh does or the large language models perform better with the direct comparison than they do with the more complex demand for uh metalinguistic judgment so the fact that the model has to process the numbering of the uh options and then answer in terms of a number uh is a subtle but nevertheless um important additional variable in the experiment and that can influence the model's capacity to get the answer right Rafael do you want to add to that no I think that well maybe we can mention briefly the the order example we discuss which is from um this paper by Andre Lan um which which kind of um has a little extra ingredient that makes the example interesting and even more problem in terms of comparative psychology which is um one way in which auxiliary task demands can be ignored or um disregarded or uh overlooked is when uh you are doing a direct comparison between humans and LMS on a task and the experimental conditions are mismatched in such a way that the task as you set up imp the stronger demands stronger task auxiliary demands on the LM than it does on human subjects that's something that that can happen quite often and so there is this this interesting example from a couple of papers originally published by black rights and colleagues from Stand the H group where they looked at um the ability of language models to handle recursion um looking at Center B it closes uh how such closes might um for example when you add a prepositional phrase with the subject

of the sentence and the verb might throw off either humans or llms into agreeing the the verb in the wrong way so giving the wrong number to the verb um for example the keys that the man put on the table here it should be R because Keys is plural but because you have closes in the middle uh and if you add more closes like this that un bedded in the middle people and alms can get confused and and U predict that the verb should be EAS for example so um they tested this on humans nms and found that on the more complex examples involving complex more complex constructions or recursion um humans were doing decently well but performance was collapsing compared to the simpler examples and and from Deep Mind um looked into that and realized that the experimental conditions were mismatched so the humans as is very common in in ctive science experiments were getting some training before they completed the test items to just get familiarized with the task so they they were given some examples of the task um for each condition and the LM were just prompted zero shot as um people usually put it so just um without any example just on blank and uh Andrew found out that if you he he he replicated the experiments but added some proper matched uh testing conditions for the LM um so adding some examples of the task in the prompt what it's known as F shot prompting and with that he found that performance was equivalent in in fact the LM that he tested was slightly better on a more complex constructions than humans so when you match the test conditions here you actually match also I you it's it's not it's not automatic but you you you can match the test demands I mean it could be that there are reasons why the very same experimental conditions would result in different demands for humans and other but you still in this case even out the playing Ground playing field uh in such a way that you don't find the behavioral discrepancy that you found ini any um yeah um good so shall we continue to the next yeah section um so another uh another way that auxiliary or another type of auxiliary task demand is input independent computational limitations um and here we're thinking of a few papers that show that the number of forward passes that a Transformer can make uh influences its ability to find the right spot in parameter space so you neural networks are um function approximators uh but they um their ability to approximate a function can be eliminated uh can be limited by the um the the number of computations it's allowed to reform and the sort of crucial feature of uh Transformers in this example is that um the number of operations that determines the next token is limited by the number number of tokens that it's seen so far um and it turns out that um if you train a Transformer with um additional meaningless tokens um like pause tokens like the word pause uh you can increase its accuracy across a range of of question typ and yeah this is this counts as an auxiliary task demand in our

view because um it's doing something roughly analogous to sort of giving the model that's demanded auxiliary Factor right but not yeah yeah sorry I'll yeah um it's it's doing something like giving the model uh time to think and um yeah so so you might think that the absence of that additional inference time is uh a factor that is not directly not conceptually related to its capacity to answer uh a question like the one on the screen um you know a simple arithmetical question uh ref do you want to fill in more yeah yeah no so so I think that analogy is a nice one time to think because if you if you tested a human on even simple mathematical questions or any any task really and just as them you know tell them they have like one second to just blur out an answer performance would probably be pretty bad um and you can think of asking an llm to answer a question point blank as very very loely analogous to that obviously this is an analogy and there are very important differences here but I think it's a helpful heuristic to think about what is um what is going on roughly here namely uh in in above cases the system the human or the llm does not get the chance to perform the necessary computations to derive the correct answer and so yeah what we talk about in the paper is that you have this experimental work showing that if you ask a question to language model um the amount of tokens it's a it's it's allowed to generate before providing the answer um makes a difference to how accurate it is so if if uh it just generates a few tokens then have to give an answer or even if it just has to give the answer point blank with the very first token it's going to be less accurate that if you give it a chance to generate a number of tokens before giving the answer so the usual way in which this is understood is that when you ask when you you you allow the LM to generate a number of tokens before getting the answer or you even prompted to do so you say think step by step for example um that's known as Chain of Thought prompting and essentially what you're doing is you're forcing the llm to generate a reasoning Trace what looks outwardly externally like a reasoning Trace in the output before giving an answer and we know CH of thought prompting increases performance accuracy um but what was found by a couple of papers um is that the uh mechanistic influence of this process is not entirely due to the nature of the tokens that are generated in this reasoning Trace in other words it's not just that the llm has to um generate the right tokens corresponding to different steps of reasoning before giving an answer in fact the very fact that you allow the llm to just generate tokens any token before giving an answer from a mechanistic perspective affords the system to perform additional computations that can complete a computational circuit that otherwise wouldn't get a chance to be completed and to derive the correct answer so as Charles mentioned you can have you can set up an experiment when you have the llm just generate meaningless tokens like Lal

just dots a bunch of dots dot tokens before giving the answer and the more dots you allow before the token gives the answer the greater the expressivity of the system and the more um the more kinds of problem problems you can answer um and so as Charles mentioned every time an LM is generating a token the LM is performing one forward pass um and so the more tokens it's generating the more forward passes it's doing and one way to think about what's going on here as well is that having these additional forward passes where you you fit back the whole input sequence plus the previously generated tokens into the system to generate the next is also a way to introduce the form of recurrence in Transformers that are not in terms of the architecture recurrent networks um so that increases the facility and uh you know in complexity their terms and uh yeah there is there is pretty compelling evidence that um if you don't allow for that then you're imposing a limitation that again we think is very very loosely analogous to prompting a human to Point Long question without thinking um so that's an S an auxiliary Factor because if you give the LM the opportunity to generate enough tokens um it might have the competence to um solve a task but you might not see that otherwise and you might get from errors that you to think it's compelling yeah okay um so the the third type of type one anthropocentric U bias that we talk about is mechanistic interference and so um this comes from the mechanistic interpretability work and the basic idea is that because um large language models are capable of in context learning they can uh learn different strategies for solving a different particular type of problem and the strategy that they Implement at a given time can be different so um you can talk about this in terms of uh virtual circuits that are formed inside the language model and um there's some interesting work from uh Neil Nanda and others uh showing that in some circumstances uh these two circuits can compete with one another um so at a certain level of uh or after a certain amount of training um you get one circuit operative after a bit more training you have two different circuits um but they're uh the first circuit is still sort of dominant and um then after additional training the model um converges on on the second circuit and uh the first one uh slowly gets um sort of it sort of ceases to influence the internal operations of the model and it's only once you reach that third uh phase uh at which the the benefits of the second circuit with respect to the first become visible so you can you can show using decoding work that the second circuit is there uh before you can show behaviorally that the second circuit uh yields better performance accuracy on on the task so um I I suppose there's a combination of two ideas here one is that um there are different strategies a model can Implement for solving a problem we can detect those strategies internally using decoding methods um so three ideas and then the third is uh a good

strategy can be uh present in some sense in the model um before it has had the chance to influence behavior um and and so this is just another way that the link between performance and competence is shown to be complicated than it might seem at first raap can certainly exp and um yeah just to to clarify one thing so the circuits are just um you know to think about the caal structure of new network and essentially computational subgraphs of the network that have a specific function you can think of a circuit as implementing a particular algorithm or set of computations um and so part of what people are interested in in this mechanistic inter literature that we build on people like neiland Chola and others is reverse engineering the circuits that in deep networks and large language models um peer to implement certain well well defined algorithms in some cases at least um and the emerging picture that we build on here is that um there is a lot of redundancy built into neuron networks as they learn to U perform a task optimizes function that in many cases translate into redundant circuits that relate to the same tasks the same kinds of um the same kinds of input output mappings and uh for these circuits might be somewhat identical circuits that are just redundant or they might be different algorithm to to to do similar thing and different strategies to solve some problem as where maybe one will be a bit more approximative and the other one a bit more exact more computationally intensive so that's where you can have some uh interference um where one um or at Le some competition where one circuit takes over another and um such that the other becomes kind of you know it's it's there it's latent in the system but doesn't get a chance to inflence behavior on a specific input so you can get a performance error for that reason and this can combine with the other things we mentioned here so things like Tas demands the first thing we discussed as well as the number of tokens you generate both of these things could cause a particle of circuit to take over another um so it's it's we can think of this holistically um perhaps if you ask a question point blank to a model without letting it generate a bunch of tokens before getting an answer then one particular approximate circuit might take over that gives the wrong answer if you let it generate more tokens then another morea circuit might be given a chance to um in C inflence the output reasing in the right answer and similarly we task demands uh strong task demands might uh in some cases um impede on the uh triggering of a certain circuit that would otherwise have given the right answer um so that could be the case perhaps in the LCR and linan example where giving examples of the task in the PRS might actually Prime the right circuit to solve the task about complex recursive cases in the right way um so yeah these are the three main auxiliary factors that would relate to what we call type one anthropomorphism anism sorry I guess we should we been AIT long already so we should

be quick on type two do you want to briefly explain type two child yeah so type type one uh deals with cases where performance of the model is um weak compared to humans so the model doesn't do so well um and then type two is when the model does do well but nevertheless is different in some respect from the uh performance profile of the human or we have some evidence to think that the model uses a different strategy than humans typically use and um the idea is that um even once you hold performance equivalent or average performance equivalent um you know making a different pattern of errors or adopting a different strategy as evidenced by uh you know some interpretability work any deviance from the human strategy is evidence of fragility or only uh a trick solution um and uh this point is a bit more philosophical I suppose but the um idea is that the human strategy for solving a problem um isn't necessarily the most General strategy for solving a problem and uh what matters is whether the strategy that is pursued by the model is General whether it's robust whether it's accurate uh and not merely whether it mirrors the human strategy um yeah and we we end the we end the behavior by considering an objection um which is um why like given that um in humans we study cognition largely through language um and given that LMS are trained on language or on linguistic outputs from humans um isn't it appropriate after role to treat um human cognition as the correct or appropriate thetic to study LMS and we to that we answer that it depends how we think of of of that dialectic um so we acknowledge that there is there is no really other option than to start our investigation of cognitive abilities in llms by with reference to human cognitive abilities using human cognitive abilities as some kind of real reference points things like FY of Mind memory metacognition R forms of reasoning Etc that are familiar to us because we humans have them um and this is the same thing by the way in animal cognition for example or in developmental psychology where in any comparative psychology setup um the reference point for what concept psychological capacities initially at least um is is necessarily tied up with our conception of what we human we humans have in our repertoire of capacities but we emphasize that this is only the starting point so here we B ter from the philosopher Ali Bo who calls this investigative kinds investigative cognitive kinds we start with a cognitive kind like memory or metacognition tic memory met the of mind um as as an investigative starting point the starting point to the investigation and then we have we we can try start operational operationalizing this concept this kind this cognitive capacity in an experiment testing LMS on it with an open minded empirical approach and then based on the results from that iteratively refine the capacities that we are T the capacity that we're targeting or definition of the capacity we're targeting in a way

that could gradually lead us to shed the the initial anthropocentric assumptions that we have such that as the experimental um project runs a COR or as as uh we make we get more results and and refine our Concepts we may end up with um something that no longer looks like looking trying to find human like memory in Ravens or uh in um or in LMS but ends up looking like looking for something that some capacity that is that shares some similarity with human human episodic memory but is different in other respects um and so we can gradually come up with a kind of cognitive ontology for the systems that is less anthropocentric um so we emphasize that this is kind of dows feedback look here that's that's that premise on open-minded empirical investigation that doesn't settle this questions a priori but has to start as a necessary starting point with the the reference to human cognition I don't know if you want to add to that J um no I think that's pretty good maybe we should uh move on to questions yeah sound good awesome wow you should stop sharing my screen uh you you can stop sharing or you could leave it up but I can move it awesome okay yeah a lot of interesting uh pieces there so thank you I'll read some questions from the live chat but first I just wanted to read a short quote from the 2022 active inference textbook they wrote um on page 195 some decades ago the philosopher dennit lamented that cognitive science scientists devote too much effort to modeling isolated subsystems EG perception language understanding whose boundaries are often arbitrary he suggested to try modeling the whole iguana a complete cognitive creature perhaps a simple one and environmental Niche for it to cope with so it's interesting about the approach that you're taking this is kind of a simple synthetic iguana but that's leading to to to the the bringing to Bear of a lot of these empirical phenomena because there is something and and and so I saw in the presentation paper kind of this call for like deliberate investigation rather than just chopping up the Iguana a priori with a framework that that applies to humans or that centers humans or or that just uh soothes the epistemic challenge that's presented okay okay first question from Dave he wrote have you looked at Daniel dennett's distinction between competence without awareness and competence with awareness he expands on this in the 2023 from bacteria to B and back I find this much more valuable than chomsky's highly problematic performance without competence a situation that chsky posits but doesn't look at deeply where do you put awareness in all of this competency uh well maybe I'll let you take uh that one in a second CH because you're you're maybe more than than I am but I'll just say um awareness is a very Pocus term like many terms uh in philosophy L but particularly this one more than many others I think so um it can mean lot of different things all different contexts um here we don't focus on things like Consciousness because I think we probably

both agree that it's a less tractable uh maybe empirical problem to try to assess the presence or absence of Consciousness in language models even though many people are interested in that uh we think that we have more hope of making progress in the near term with more well defined um cognitive capacities or C functions and things that relate to forms of certain forms of reasoning variable binding uh Etc um so we our framework in principle would apply to things like Consciousness or as you put awareness generally speaking but uh we don't really focus on that for examples the the other quick thing I wanted to mention is I I seem to remember that the frame from Den but again I'm not a denit scholar was competence without comprehension um uh which is seems a little different from competence without awareness perhaps depending on how you think of comprehension um and yeah I think that does have I think it is a very interesting phrase that it does um in fact I I have this project that un published with Chris Bela who I think you had on the podcast as well um on semantic competence in language models where we use that phrase um to kind of avoid taking its stance on the kind of messy Murdy question of whether llms understand language uh which builds in all sorts of assumption including about Consciousness actually for some people like johno um and we focus on a more restricted notion of competence and I think our paper here also has that property that we we operationalize competence we end up operationalizing competence in terms of the sets of knowledge of mechanism and mechanisms that enable a system to generalize well in a given domain basically and in a way that's I suppose fairly deflationary compared to some more expensive understandings of competence that are related to comprehension or understanding more broadly but I'll let you take back Charles no yeah that was that was good um the phrase yeah the distinction that Den draws is between competence with comprehension and without and I think um competence with comprehension is the ability not just to pursue a strategy that's successful for solving a problem but to um articulate the strategy in such a way that you could teach it for example and um humans only sometimes have competence with comprehension we have many competences that lack comprehension right um you know when we learn to walk for example um we have an amazing competence that we still can't quite translate into robotics because we don't fully understand how it works uh and when it comes to large language models I think we should not expect uh comprehension I mean they have an amazing Suite of competences if you thought that they also had comprehension then I suppose you would think like well if you want to understand how a large language model works you can just ask it but that's that's a bad strategy nobody nobody thinks that's a good way to find out um how a large language model works so so they're on the um competence without

comprehension side of things um and in order to figure out what in order to figure out what the mechanisms are that enable its competencies we have to pursue strategies that are broadly similar to the strategies we using you know cognitive psychology or cognitive Linguistics um and you know we have to run experiments so I think that that's all very compatible with Dan's way of looking at things um one other thing I'll mention um Dan's so so Dan was uh quite influential to me and we actually wrote A Commentary together which pushes back a little bit on a simple understanding of this distinction so we were looking at um the evolution of metacognition and basically what we argue is that um given the gradualism of evolution there must have been something in between base level cognition and metacognition so we shouldn't see that distinction as black and white and um you know I think that if you want to contrast the sort of uh cognitive prowess of a human adult with lots of linguistic scientific uh Concepts at her disposal with you know a non-human animal then this strong distinction between competence with and without comprehension is reasonable um but in the space of all possible Minds we should be open to the view that there can be you know semic competent um forms of cognition and just follow Qui on this uh it occur to me while I was listening to you as well that um the first example of auxiliary practice we give auxiliary test demands and specifically the Hugh and Frank example is a nice is a nice example where in order to give the metal linguistic judgment correctly succeed at that you would need confidence with comprehension because you need to understand not only be able to to to to agree the verb with the subject but know the rule and know how to formulate it perhaps reize it uh for example to teach someone right and so uh and so when you find that the LM can do well at the low Tas demand version of the task and of the high test the explicit metalinguistic judgments in some way that's an example of the having confidence with compion nice awesome okay upcycle Club wrote question given that llms inherently reflect SCT anthropocentric biases due to their training on human data and goals how can we ensure that their Intermodal discourse aligns with Humanity's values um so the they're inter what discourse sorry the inter model discourse perhaps amongst the models I see I see like in that Farmville paper yeah the small the yeah generative agents yeah the small on yeah I I think that's beyond the scope of this paper to be honest but um I mean we could mute about it yeah but I don't know that we have I don't know if this project has much I mean I think we both have an interest in the alignment problem uh independently of this project but I don't think this project has much to say really about this I don't know CH what you think uh yeah yeah and I don't yeah I don't have anything super concrete on that okay Dave asks an example of inserting noise into llm training that was the

section about the extra tokens do you see any analog to intermittent reinforced to uncertainty tolerance because you mentioned the extra tokens in the Chain of Thought and how that could also be replaced by by dot dot dot dot dot and so like what is that telling us about model training when um it seems like there's some situations where adding Superfluous tokens would diminish signal in data sets but then here are other situations where it seems to actually help um um yeah so in that particular paper I think it's called thinking dot by Dot and there is a subtitle um it's by Will Mar Jacob I think um in that paper um if I correctly what they did is that they they introduced just this one F token it's one meaningless token just a just a DOT and train the model to give an answer after producing a certain number of dots that's not just like introducing random gibberish in your training data it's actually quite a specific intervention that forces the model to um learn to perform certain computations before giving an answer um so so it makes sense to me that this couldn't diminish performance and you could get benefit from that it's not quite the same as just having that training data right um just because the token seems meaningless it's a f token it's a DOT um it's not just random gibberish is just going to throw off the model and and impede its uh the optimization of its learning function or or this good Downstream performance um but what it's going to do is going to force the model to learn that when there is a DOT token it can allocate computation with his attention head and other parts of the architecture in such a way that it's um getting towards deriving the correct token when it's finally producing the token that matters and that's meaningful after the series of thoughts um yeah I don't know Charles if you have another answers now yeah I mean I mean I think it's an important question because a priori uh if someone said look we're going to append and prepend a whole bunch of meaningless symbols to an LLM input you might very well think that this will just weaken the signal to noise ratio and degrade model performance so it's against that background that the empirical result that it doesn't degrade model performance um ought to be regarded as an important clue about how the model Works um so so so I think that the intuition behind this question is indeed part of the interpretation of the empirical results right it's it's surprising for exactly this reason and then um the theory that's supposed to uh you know well this is an active inference podcast right so the theory that's supposed to help rid some of the surprise here is um the idea that given the uh architecture of a transformer where it's it has to go through all the tokens in every cycle um having these extra gives it uh sort of more computational bandwidth and therefore more expressivity or more capacity to uh you know locate uh the right spot and parameter space and and even that in a way reminds me of let's so so it's not just that dots improve performance it's not it's that it was like you mentioned

it was trained to have that and similarly it it could have been trained maybe hypothetically to just output Shakespeare quotes verbatim while you're processing so that's kind of like a filler or more of like a sort of that was a great question it's like these are linguistic paddings that that do create time to to get to the meet so not only does it signal and signpost if it's being trained to to have that meaning which then questions like so then was it a meaningless dot if it if it had a um a cognitive or even like a a semantic um aspect I had a question how do you feel like in this era Cambrian explosion of diverse intelligences how can we understand capacities when they seem so conditional upon the setting and how the system of Interest is interacted with like what are the practical implications for people who are studying llms and other synthetic intelligences from like a safety or reliability or performance perspective you want to no I was throwing it to you so so so just want make sure the question so the question is how the how is the notion of a capacity um changing when we have such different systems that seem to have intelligent Behavior yeah and it's so dependent upon potentially initially unintuitive ways of interacting so H how how can we understand the the reliability and the performance and the capacity of of a model other than for example by exhaustively inputting prompts which can really happen what what could we really say or no Andor just how do you feel that this work re-enters into the ways that people practically are using the models right okay yeah uh it's a very interesting question so the first part of the question is I think you know part of the background assumption from for this paper that I've explicitly defended in other work is that behavioral evidence is simply not sufficient in most cases to arbitrate disputes about capacities of LMS um when it comes to human cognition we do have to rely a lot on on psychological experiments that are ultimately behavioral and we do also r on self reports in a little more than we can when it comes to alabs because we despite the move away from relying on intuition and introspection in the history of psychology it still has a role to play but with by and large probity towards by behavioral experiments that get increasingly sophisticated um to re try to regress engineer what's going on inside the black box when it comes to LM um LMS partly because they're so different from us um relying exclusively on behavior is even more is even more difficult because we have even less of an idea of what might be going on inside the black box and whether it's anything like what's going inside our blackbox and we have reasons to think in fact it might be very different so I think I think we both agree that we have to supplement this with mechanistic work that essentially involve performing causal interventions on the inner mechanisms on the inner workings of the systems so decoding representation and computations that the systems uh that are in principle available

to the systems and then intervening on them to confirm hypotheses about the role of these representations and computations and we have methods to do that in part why we can be a little optimistic about this project even though it's extremely challenging especially to be scaled up to large models is because unlike what's happening in your Cent the brain where the range of decoding methods and intervention methods we had is extremely limited but for ethical and for simply um practical reasons we don't just don't have ground truth access to activations in neurons at least that easily and we also are generally enabled to make specific interventions on activations in the brain uh when it comes to alms we have full ground truth knowledge of all activations of every single part of the network and we also have full access to all of it for interv at inference time so that that opens up a whole new range of things we can do and that enables us to go beyond Behavioral Studies and actually decode these features and circuits or as researchers put it in the literature or as philosophers would generally put it uh representations and computations that the system is actually making use of and try to reverse engineer what kind of what kind of algorithms it's it's making use of so part of the broader problem projects that we have with Charles is to um suppose we start with the investig investigative kinds as we put as as alib calls this human C capacities um we can operationalize them and do behavioral experiment from the top down and then from the bottom up we can also try to reverse engineer the mechanic building blocks of the computations and representations that LMS make use of to solve the task related to that particular capacity T and then we can meet somewhere in the middle and try to uh from that line of work that approaches things from above ends um bring to the four some kinds of mid-level abstractions as we call it or computational building blocks that might be key to um the performance of the system in that domain so for example if you're interested in the capacity for reasoning you can start with this very broad humancentric notion of reason reasoning then try to operationalize it in a reasoning task then do some behavioral testing and then mechanic interpretability on that reasoning task find out how the system is solving it find out how the LM is doing well and why reverse engineer building blocks that might for example have to viable manipulation viable binding and then from there you might be able to iteratively refine the notion of reasoning you started with uh to have a more specific and per less human C notion that is now operationalized in lowlevel um More Level terms like uh you know that that that involve the manipulation of variables in certain ways and the Banning of variables to figures Etc um so yeah so that's I think the general approach we take now how that does that does any of that Feit back into interactions how we humans interact with LMS I think perhaps one way in

which it could feedback is simply um in terms of challenging or our um spontaneous anthropomorphic attitudes TOS uh to some extent the same way if you read a lot about Ang cognition perhaps you will interact with your cat in a slightly different way uh that you might maybe not rush the conclusion that uh when your cat performs a certain Behavior it has understood what you think and it's modeling what you're thinking about what it's thinking or something um perhaps you might um adopt a more deflationary attitude to uh explain the behavior of your cat doesn't mean you have to love them any less or doesn't mean you have to uh um you know if there is the other thing like if you want at the end of the day to speak to your cat like a human because you the r Gman for that then that's you know all the more part to you and the same way if you find it useful to treat llms in the way you interact with them um to to have F interactions with them to treat them as if they had beliefs designs Etc Al of human like capacities then that's fine if that's for actual purposes but at least if that line of work that we are kind of sketching here ends up maturing enough the hope is that we can interact with LMS perhaps in a way that's um where even if we if we have that kind of intentional stance and make believe about what the kind capacities they have at the background we will know that uh what their limitations are and what their actual capacities are I don't know if you agree with that Charles because we didn't discuss that much yeah yeah no I agree with all that I just had a slightly different um first reaction to the question I took the question to be in part about how to deal with the sort of prompt sensitivity of models the fact that sometimes we you know write something that seems natural to us but provokes an unexpected response from a large language model and how do we think about that um and and the first thought that occurred to me was just that um we should distinguish between different kinds of large language models um you know we have the sort of uh huge large language models which are um fine-tuned to interact with us in a particular way and are and here's the central point they're trained on a a sort of unthinkably large database whereas um there are other sorts of large language models where the training data is more circumscribed and where we know uh in more detail you know what um where you can survey uh what the training data says and I think if you're interested in you know what the mechanisms are underlying the responses um it's certainly very helpful to uh look at smaller but nevertheless large language models where the training data is known to us because um you know when you train a model on the entire internet there are going to be all kinds of uh you know subtle signals in there that we don't have much hope of tracing back to their Source but which will influence the Model Behavior in all sorts of ways um but working with these somewhat more conscripted uh

models uh gets rid of that problem at least in part cool well where where do you see the work going or where do you plan to continue this direction yeah so we actually so we wrote this paper this short paper for uh the icml machine learning conference International conference machine learning um that's happening this week in Vienna and we'll be um getting to Vienna at the end of the week for the particular workshop at which we're presenting this uh paper which is a workshop on language models and cognitive science um so there is a very strict page limit for the ZL contribution which is four pages but what we want to do next is to um expand this into a more philosophically substantive paper that's going to be a bit longer um and that's going to expand on the more philosophically mey parts of that of that project um because everything is a bit compressed in that version that we're presenting on HML so yeah that's the next step for us this is a really useful way first to force ourselves to write things down after running the Box piece we wanted to write an academic piece now we've written kind of the condens skeleton of the piece that focuses more that c caters more to an ml audience and now the next step is to write the full philosophy paper or at least that part of the project to be complete I don't have to that people have ideas yeah yeah I got nothing to add to that cool yeah it's well it's very interesting work I think it it brings a lot of pieces together and it's some philosophy and cognitive science jumping in jumping into the the Heat and into the the spotlight and the relevance and so it's going to be an exciting Learning Journey thanks for having us yes yeah thank you very much cool enjoyable conversation till next time thank bye e